

Using (11.24) and (11.25) in (11.23), the overall scattering matrix is given by

$$S_p = \begin{bmatrix} S_{11}^A & 0 \\ 0 & S_{22}^B \end{bmatrix} + \begin{bmatrix} S_{12}^A & 0 & 0 & 0 \\ 0 & 0 & S_{23}^B & S_{21}^B \end{bmatrix} \begin{bmatrix} -S_{22}^A & 0 & 0 & 1 \\ 0 & -S^C & 1 & 0 \\ 0 & 1 & -S_{33}^B & -S_{31}^B \\ 1 & 0 & -S_{13}^B & -S_{11}^B \end{bmatrix}^{-1} \begin{bmatrix} S_{21}^A & 0 \\ 0 & 0 \\ 0 & S_{32}^B \\ 0 & S_{12}^B \end{bmatrix} \quad (11.26)$$

For arbitrary inputs at the two external ports, the incoming wave variables at the internal ports can be found using (11.21), as follows:

$$\begin{bmatrix} a_2 \\ a_6 \\ a_5 \\ a_3 \end{bmatrix} = \begin{bmatrix} -S_{22}^A & 0 & 0 & 1 \\ 0 & -S^C & 1 & 0 \\ 0 & 1 & -S_{33}^B & -S_{31}^B \\ 1 & 0 & -S_{13}^B & -S_{11}^B \end{bmatrix}^{-1} \begin{bmatrix} S_{21}^A & 0 \\ 0 & 0 \\ 0 & S_{32}^B \\ 0 & S_{12}^B \end{bmatrix} \begin{bmatrix} a_1 \\ a_4 \end{bmatrix} \quad (11.27)$$

The outgoing wave variables $[b_2 \ b_6 \ b_5 \ b_3]^t$ at the internal ports can be obtained by substituting the above relation in (11.25).

To obtain the S-parameters for this circuit using the method given in Section 11.2.1, we put independent generators at the external ports. The generators (shown dotted in Figure 11.11) are single port components and they are characterized by

$$b_7 = c_1 \quad (11.28a)$$

$$b_8 = c_2 \quad (11.28b)$$

- 1) Draw block diagram
- 2) Set up S matrix 11.29
- 3) Set up Γ matrix 11.30
- 4) Set up ρ -S matrix 11.31
- 5) Invert ρ -S matrix
- 6) Use [c] to solve [a] 11.21
- 7) Use [a] to solve [b] 11.30

Code
Start

11.2.1

MCAP

Method

(P579)

and

10/11/16

Rev

7/17/18

The relation between b's, a's and c's for all the components of the circuit can now be written in the form of (11.11) as

$$[b] = [a] + [c] \quad (11.11)$$

$$\begin{bmatrix} b_8 \\ b_7 \\ b_6 \\ b_5 \\ b_4 \\ b_3 \\ b_2 \\ b_1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_8 \\ a_7 \\ a_6 \\ a_5 \\ a_4 \\ a_3 \\ a_2 \\ a_1 \end{bmatrix} + \begin{bmatrix} c_2 \\ c_1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

(11.29)

The interconnection matrix $[I]$ now becomes

$$[b] = [I][a]$$

$$\begin{bmatrix} b_8 \\ b_7 \\ b_6 \\ b_5 \\ b_4 \\ b_3 \\ b_2 \\ b_1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_8 \\ a_7 \\ a_6 \\ a_5 \\ a_4 \\ a_3 \\ a_2 \\ a_1 \end{bmatrix}$$

(11.30)

$$[b] = [I][a]$$

11.14

The W matrix must be Inverted!

Evaluation of Circuit Performance

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Using (11.17), the incoming wave variables at the ports are determined as

(11.17)

$$[a] = [W = \Gamma - S]^{-1} [c]$$

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \\ a_7 \\ a_8 \end{bmatrix} = \begin{bmatrix} -S_{11}^A & -S_{12}^A & 0 & 0 & 0 & 0 & 1 & 0 \\ -S_{21}^A & -S_{22}^A & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -S_{11}^B & -S_{12}^B & -S_{13}^B & 0 & 0 & 0 \\ 0 & 0 & -S_{21}^B & -S_{22}^B & -S_{23}^B & 0 & 0 & 1 \\ 0 & 0 & -S_{31}^B & -S_{32}^B & -S_{33}^B & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -S^C & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & S_{11} & S_{12} \\ 0 & 0 & 0 & 1 & 0 & 0 & S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ c_1 \\ c_2 \end{bmatrix}$$

Final cascaded S parameters!

(11.31)

and the outgoing wave variables at all the ports, b , can be determined using (11.30) and (11.31).

The values of the wave variables at all the ports can thus be obtained for any combination of values of c_1 and c_2 . If $c_1 = 1$ and $c_2 = 0$, the values of b_1 and b_4 (at the external ports) correspond to S_{11} and S_{21} , respectively, of the overall scattering matrix. Similarly, for $c_1 = 0$ and $c_2 = 1$, the parameters S_{12} and S_{22} correspond to the values b_1 and b_4 , respectively.

codex
end

11.2.4 Analysis by Subnetwork Growth Method [6]

Using the technique described in Section 11.2.2 to obtain the S-matrix of a network with multiport components, a matrix of order equal to the number of interconnected ports must be inverted. When the network contains many interconnected ports, the order of the matrix to be inverted will become quite large. The computational effort of inverting the matrix can be reduced considerably if the entire network is not taken at once, but is partitioned into a number of subnetworks. The S-matrices of the subnetworks are obtained separately